MSG No. 226.122 $Fm\bar{3}c$ [Type I, cubic]

Table 1: Wyckoff site: 8a, site symmetry: 432

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,12,14,16,17,18,53,54,56,61,70,71,98,103,105,111,115,120,147,148,154,155,164,165]
2	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[2,7,9,15,19,24,51,52,58,59,68,69,97,108,110,112,113,114,149,150,152,157,166,167]
3	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[3,4,10,11,20,21,50,55,57,63,67,72,101,102,104,109,118,119,145,156,158,160,161,162]
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[5,6,8,13,22,23,49,60,62,64,65,66,99,100,106,107,116,117,146,151,153,159,163,168]
5	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[25,36,38,40,41,42,77,78,80,85,94,95,122,127,129,135,139,144,171,172,178,179,188,189]
6	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[26,31,33,39,43,48,75,76,82,83,92,93,121,132,134,136,137,138,173,174,176,181,190,191]
7	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[27,28,34,35,44,45,74,79,81,87,91,96,125,126,128,133,142,143,169,180,182,184,185,186]
8	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[29,30,32,37,46,47,73,84,86,88,89,90,123,124,130,131,140,141,170,175,177,183,187,192]

Table 2: Wyckoff site: 8b, site symmetry: m-3.

No.	position	mapping
1	$[0, 0, 0]$	[1,8,9,10,17,18,19,20,21,22,23,24,25,32,33,34,41,42,43,44,45,46,47,48]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[2,3,4,5,6,7,11,12,13,14,15,16,26,27,28,29,30,31,35,36,37,38,39,40]
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[49,56,57,58,65,66,67,68,69,70,71,72,73,80,81,82,89,90,91,92,93,94,95,96]
4	$\left[\frac{1}{2}, 0, 0\right]$	[50,51,52,53,54,55,59,60,61,62,63,64,74,75,76,77,78,79,83,84,85,86,87,88]
5	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[97,104,105,106,113,114,115,116,117,118,119,120,121,128,129,130,137,138,139,140,141,142,143,144]
6	$\left[0, \frac{1}{2}, 0\right]$	[98,99,100,101,102,103,107,108,109,110,111,112,122,123,124,125,126,127,131,132,133,134,135,136]
7	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[145,152,153,154,161,162,163,164,165,166,167,168,169,176,177,178,185,186,187,188,189,190,191,192]
8	$\left[0, 0, \frac{1}{2}\right]$	[146,147,148,149,150,151,155,156,157,158,159,160,170,171,172,173,174,175,179,180,181,182,183,184]

Table 3: Wyckoff site: 24c, site symmetry: -4m.2

No.	position	mapping
1	$\left[\frac{1}{4}, 0, 0\right]$	[1,8,33,34,61,62,74,75]
2	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{2}\right]$	[2,3,37,38,57,58,73,80]
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	[4,16,29,39,162,165,187,191]
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	[5,15,28,40,163,167,186,189]
5	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{2}\right]$	[6,11,31,36,116,118,137,144]
6	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2}\right]$	[7,12,30,35,113,120,140,142]
7	$\left[\frac{3}{4}, 0, 0\right]$	[9,10,25,32,50,51,85,86]
8	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{2}\right]$	[13,14,26,27,49,56,81,82]
9	$\left[0, \frac{1}{4}, 0\right]$	[17,24,44,46,103,108,126,131]
10	$\left[0, 0, \frac{1}{4}\right]$	[18,21,43,47,148,160,173,183]
11	$\left[0, 0, \frac{3}{4}\right]$	[19,23,42,45,149,159,172,184]
12	$\left[0, \frac{3}{4}, 0\right]$	[20,22,41,48,102,107,127,132]
13	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[52,64,77,87,114,117,139,143]
14	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[53,63,76,88,115,119,138,141]
15	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	[54,59,79,84,164,166,185,192]
16	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	[55,60,78,83,161,168,188,190]
17	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	[65,72,92,94,151,156,174,179]
18	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[66,69,91,95,100,112,125,135]

continued ...

Table 3

No.	position	mapping
19	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[67,71,90,93,101,111,124,136]
20	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	[68,70,89,96,150,155,175,180]
21	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	[97,104,129,130,157,158,170,171]
22	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	[98,99,133,134,153,154,169,176]
23	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	[105,106,121,128,146,147,181,182]
24	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	[109,110,122,123,145,152,177,178]

Table 4: Wyckoff site: 24d, site symmetry: 4/m..

No.	position	mapping
1	$\left[0, \frac{1}{4}, \frac{1}{4}\right]$	[1,32,56,73,98,123,147,170]
2	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[2,27,51,74,97,128,152,169]
3	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[3,26,50,75,104,121,145,176]
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[4,40,67,95,119,139,160,172]
5	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[5,39,66,93,117,138,159,173]
6	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[6,35,65,96,107,126,168,185]
7	$\left[\frac{3}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[7,36,68,94,108,127,166,188]
8	$\left[0, \frac{3}{4}, \frac{3}{4}\right]$	[8,25,49,80,99,122,146,171]
9	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	[9,34,58,81,110,133,157,182]
10	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	[10,33,57,82,109,134,158,181]
11	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[11,30,72,89,102,131,161,192]
12	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[12,31,70,92,103,132,164,190]
13	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[13,38,62,85,106,129,153,178]
14	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$	[14,37,61,86,105,130,154,177]
15	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[15,29,69,90,114,141,149,183]
16	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[16,28,71,91,115,143,148,184]
17	$\left[\frac{1}{4}, 0, \frac{1}{4}\right]$	[17,48,54,83,120,137,155,174]
18	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	[18,45,53,87,111,125,165,186]
19	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	[19,47,52,88,112,124,167,187]
20	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	[20,46,55,84,118,140,156,175]
21	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	[21,42,63,77,101,135,162,189]
22	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	[22,44,60,79,116,142,151,180]
23	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	[23,43,64,76,100,136,163,191]
24	$\left[\frac{3}{4}, 0, \frac{3}{4}\right]$	[24,41,59,78,113,144,150,179]

Table 5: Wyckoff site: 48e, site symmetry: 2mm..

No.	position	mapping
1	$[x, 0, 0]$	[1,8,33,34]
2	$\left[x + \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[2,3,37,38]
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - x\right]$	[4,16,29,39]
4	$\left[\frac{1}{2}, \frac{1}{2}, x + \frac{1}{2}\right]$	[5,15,28,40]
5	$\left[\frac{1}{2}, x + \frac{1}{2}, \frac{1}{2}\right]$	[6,11,31,36]
6	$\left[\frac{1}{2}, \frac{1}{2} - x, \frac{1}{2}\right]$	[7,12,30,35]

continued ...

Table 5

No.	position	mapping
7	$[-x,\ 0,\ 0]$	[9,10,25,32]
8	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[13,14,26,27]
9	$[0,\ x,\ 0]$	[17,24,44,46]
10	$[0,\ 0,\ x]$	[18,21,43,47]
11	$[0,\ 0,\ -x]$	[19,23,42,45]
12	$[0,\ -x,\ 0]$	[20,22,41,48]
13	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[49,56,81,82]
14	$\left[x+\frac{1}{2},\ 0,\ 0\right]$	[50,51,85,86]
15	$\left[\frac{1}{2},\ 0,\ -x\right]$	[52,64,77,87]
16	$\left[\frac{1}{2},\ 0,\ x\right]$	[53,63,76,88]
17	$\left[\frac{1}{2},\ x,\ 0\right]$	[54,59,79,84]
18	$\left[\frac{1}{2},\ -x,\ 0\right]$	[55,60,78,83]
19	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[57,58,73,80]
20	$\left[\frac{1}{2}-x,\ 0,\ 0\right]$	[61,62,74,75]
21	$\left[0,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[65,72,92,94]
22	$\left[0,\ \frac{1}{2},\ x+\frac{1}{2}\right]$	[66,69,91,95]
23	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-x\right]$	[67,71,90,93]
24	$\left[0,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[68,70,89,96]
25	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[97,104,129,130]
26	$\left[x,\ \frac{1}{2},\ 0\right]$	[98,99,133,134]
27	$\left[0,\ \frac{1}{2},\ -x\right]$	[100,112,125,135]
28	$\left[0,\ \frac{1}{2},\ x\right]$	[101,111,124,136]
29	$\left[0,\ x+\frac{1}{2},\ 0\right]$	[102,107,127,132]
30	$\left[0,\ \frac{1}{2}-x,\ 0\right]$	[103,108,126,131]
31	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	[105,106,121,128]
32	$\left[-x,\ \frac{1}{2},\ 0\right]$	[109,110,122,123]
33	$\left[\frac{1}{2},\ x,\ \frac{1}{2}\right]$	[113,120,140,142]
34	$\left[\frac{1}{2},\ 0,\ x+\frac{1}{2}\right]$	[114,117,139,143]
35	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-x\right]$	[115,119,138,141]
36	$\left[\frac{1}{2},\ -x,\ \frac{1}{2}\right]$	[116,118,137,144]
37	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[145,152,177,178]
38	$\left[x,\ 0,\ \frac{1}{2}\right]$	[146,147,181,182]
39	$\left[0,\ 0,\ \frac{1}{2}-x\right]$	[148,160,173,183]
40	$\left[0,\ 0,\ x+\frac{1}{2}\right]$	[149,159,172,184]
41	$\left[0,\ x,\ \frac{1}{2}\right]$	[150,155,175,180]
42	$\left[0,\ -x,\ \frac{1}{2}\right]$	[151,156,174,179]
43	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	[153,154,169,176]
44	$\left[-x,\ 0,\ \frac{1}{2}\right]$	[157,158,170,171]
45	$\left[\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	[161,168,188,190]
46	$\left[\frac{1}{2},\ \frac{1}{2},\ x\right]$	[162,165,187,191]
47	$\left[\frac{1}{2},\ \frac{1}{2},\ -x\right]$	[163,167,186,189]
48	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	[164,166,185,192]

Table 6: Wyckoff site: `48f`, site symmetry: `4..`

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,56,98,147]`
2	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[2,51,97,152]`
3	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[3,50,104,145]`
4	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}-x\right]$	`[4,67,119,160]`
5	$\left[\frac{1}{4},\ \frac{3}{4},\ x+\frac{1}{2}\right]$	`[5,66,117,159]`
6	$\left[\frac{1}{4},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	`[6,65,107,168]`
7	$\left[\frac{3}{4},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[7,68,108,166]`
8	$\left[x,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[8,49,99,146]`
9	$\left[-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[9,58,110,157]`
10	$\left[-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[10,57,109,158]`
11	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[11,72,102,161]`
12	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	`[12,70,103,164]`
13	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[13,62,106,153]`
14	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[14,61,105,154]`
15	$\left[\frac{3}{4},\ \frac{1}{4},\ x+\frac{1}{2}\right]$	`[15,69,114,149]`
16	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}-x\right]$	`[16,71,115,148]`
17	$\left[\frac{1}{4},\ x,\ \frac{1}{4}\right]$	`[17,54,120,155]`
18	$\left[\frac{1}{4},\ \frac{1}{4},\ x\right]$	`[18,53,111,165]`
19	$\left[\frac{3}{4},\ \frac{1}{4},\ -x\right]$	`[19,52,112,167]`
20	$\left[\frac{3}{4},\ -x,\ \frac{1}{4}\right]$	`[20,55,118,156]`
21	$\left[\frac{3}{4},\ \frac{3}{4},\ x\right]$	`[21,63,101,162]`
22	$\left[\frac{1}{4},\ -x,\ \frac{3}{4}\right]$	`[22,60,116,151]`
23	$\left[\frac{1}{4},\ \frac{3}{4},\ -x\right]$	`[23,64,100,163]`
24	$\left[\frac{3}{4},\ x,\ \frac{3}{4}\right]$	`[24,59,113,150]`
25	$\left[-x,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[25,80,122,171]`
26	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[26,75,121,176]`
27	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[27,74,128,169]`
28	$\left[\frac{1}{4},\ \frac{1}{4},\ x+\frac{1}{2}\right]$	`[28,91,143,184]`
29	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-x\right]$	`[29,90,141,183]`
30	$\left[\frac{3}{4},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	`[30,89,131,192]`
31	$\left[\frac{1}{4},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[31,92,132,190]`
32	$\left[-x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[32,73,123,170]`
33	$\left[x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[33,82,134,181]`
34	$\left[x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[34,81,133,182]`
35	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[35,96,126,185]`
36	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	`[36,94,127,188]`
37	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[37,86,130,177]`
38	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[38,85,129,178]`
39	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-x\right]$	`[39,93,138,173]`
40	$\left[\frac{3}{4},\ \frac{3}{4},\ x+\frac{1}{2}\right]$	`[40,95,139,172]`
41	$\left[\frac{3}{4},\ -x,\ \frac{3}{4}\right]$	`[41,78,144,179]`
42	$\left[\frac{3}{4},\ \frac{3}{4},\ -x\right]$	`[42,77,135,189]`
43	$\left[\frac{1}{4},\ \frac{3}{4},\ x\right]$	`[43,76,136,191]`
44	$\left[\frac{1}{4},\ x,\ \frac{3}{4}\right]$	`[44,79,142,180]`
45	$\left[\frac{1}{4},\ \frac{1}{4},\ -x\right]$	`[45,87,125,186]`
46	$\left[\frac{3}{4},\ x,\ \frac{1}{4}\right]$	`[46,84,140,175]`

continued ...

Table 6

No.	position	mapping
47	$\left[\frac{3}{4}, \frac{1}{4}, x\right]$	[47,88,124,187]
48	$\left[\frac{1}{4}, -x, \frac{1}{4}\right]$	[48,83,137,174]

Table 7: Wyckoff site: 64g, site symmetry: .3.

No.	position	mapping
1	$[x, x, x]$	[1,17,18]
2	$\left[x+\frac{1}{2}, \frac{1}{2}-x, x+\frac{1}{2}\right]$	[2,7,15]
3	$\left[x+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}-x\right]$	[3,4,11]
4	$\left[\frac{1}{2}-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[5,6,13]
5	$[x, -x, -x]$	[8,22,23]
6	$[-x, x, -x]$	[9,19,24]
7	$[-x, -x, x]$	[10,20,21]
8	$\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[12,14,16]
9	$[-x, -x, -x]$	[25,41,42]
10	$\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{2}-x\right]$	[26,31,39]
11	$\left[\frac{1}{2}-x, \frac{1}{2}-x, x+\frac{1}{2}\right]$	[27,28,35]
12	$\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[29,30,37]
13	$[-x, x, x]$	[32,46,47]
14	$[x, -x, x]$	[33,43,48]
15	$[x, x, -x]$	[34,44,45]
16	$\left[x+\frac{1}{2}, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[36,38,40]
17	$\left[x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[49,65,66]
18	$\left[x+\frac{1}{2}, -x, x\right]$	[50,55,63]
19	$\left[x+\frac{1}{2}, x, -x\right]$	[51,52,59]
20	$\left[\frac{1}{2}-x, x, x\right]$	[53,54,61]
21	$\left[x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[56,70,71]
22	$\left[-x, x+\frac{1}{2}, \frac{1}{2}-x\right]$	[57,67,72]
23	$\left[-x, \frac{1}{2}-x, x+\frac{1}{2}\right]$	[58,68,69]
24	$\left[\frac{1}{2}-x, -x, -x\right]$	[60,62,64]
25	$\left[-x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[73,89,90]
26	$\left[\frac{1}{2}-x, x, -x\right]$	[74,79,87]
27	$\left[\frac{1}{2}-x, -x, x\right]$	[75,76,83]
28	$\left[x+\frac{1}{2}, -x, -x\right]$	[77,78,85]
29	$\left[-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[80,94,95]
30	$\left[x, \frac{1}{2}-x, x+\frac{1}{2}\right]$	[81,91,96]
31	$\left[x, x+\frac{1}{2}, \frac{1}{2}-x\right]$	[82,92,93]
32	$\left[x+\frac{1}{2}, x, x\right]$	[84,86,88]
33	$\left[x+\frac{1}{2}, x, x+\frac{1}{2}\right]$	[97,113,114]
34	$\left[x, \frac{1}{2}-x, x\right]$	[98,103,111]
35	$\left[x, x+\frac{1}{2}, -x\right]$	[99,100,107]
36	$\left[-x, x+\frac{1}{2}, x\right]$	[101,102,109]
37	$\left[x+\frac{1}{2}, -x, \frac{1}{2}-x\right]$	[104,118,119]
38	$\left[\frac{1}{2}-x, x, \frac{1}{2}-x\right]$	[105,115,120]
39	$\left[\frac{1}{2}-x, -x, x+\frac{1}{2}\right]$	[106,116,117]

continued ...

Table 7

No.	position	mapping
40	$\left[-x,\ \frac{1}{2}-x,\ -x\right]$	[108,110,112]
41	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}-x\right]$	[121,137,138]
42	$\left[-x,\ x+\frac{1}{2},\ -x\right]$	[122,127,135]
43	$\left[-x,\ \frac{1}{2}-x,\ x\right]$	[123,124,131]
44	$\left[x,\ \frac{1}{2}-x,\ -x\right]$	[125,126,133]
45	$\left[\frac{1}{2}-x,\ x,\ x+\frac{1}{2}\right]$	[128,142,143]
46	$\left[x+\frac{1}{2},\ -x,\ x+\frac{1}{2}\right]$	[129,139,144]
47	$\left[x+\frac{1}{2},\ x,\ \frac{1}{2}-x\right]$	[130,140,141]
48	$\left[x,\ x+\frac{1}{2},\ x\right]$	[132,134,136]
49	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ x\right]$	[145,161,162]
50	$\left[x,\ -x,\ x+\frac{1}{2}\right]$	[146,151,159]
51	$\left[x,\ x,\ \frac{1}{2}-x\right]$	[147,148,155]
52	$\left[-x,\ x,\ x+\frac{1}{2}\right]$	[149,150,157]
53	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ -x\right]$	[152,166,167]
54	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ -x\right]$	[153,163,168]
55	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ x\right]$	[154,164,165]
56	$\left[-x,\ -x,\ \frac{1}{2}-x\right]$	[156,158,160]
57	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ -x\right]$	[169,185,186]
58	$\left[-x,\ x,\ \frac{1}{2}-x\right]$	[170,175,183]
59	$\left[-x,\ -x,\ x+\frac{1}{2}\right]$	[171,172,179]
60	$\left[x,\ -x,\ \frac{1}{2}-x\right]$	[173,174,181]
61	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ x\right]$	[176,190,191]
62	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ x\right]$	[177,187,192]
63	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ -x\right]$	[178,188,189]
64	$\left[x,\ x,\ x+\frac{1}{2}\right]$	[180,182,184]

Table 8: Wyckoff site: 96h, site symmetry: ..2

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ y\right]$	[1,61]
2	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ y+\frac{1}{2}\right]$	[2,58]
3	$\left[\frac{3}{4},\ y+\frac{1}{2},\ \frac{1}{2}-y\right]$	[3,57]
4	$\left[y+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	[4,162]
5	$\left[\frac{1}{2}-y,\ y+\frac{1}{2},\ \frac{3}{4}\right]$	[5,163]
6	$\left[\frac{1}{2}-y,\ \frac{3}{4},\ y+\frac{1}{2}\right]$	[6,116]
7	$\left[y+\frac{1}{2},\ \frac{1}{4},\ y+\frac{1}{2}\right]$	[7,113]
8	$\left[\frac{1}{4},\ -y,\ -y\right]$	[8,62]
9	$\left[\frac{3}{4},\ y,\ -y\right]$	[9,51]
10	$\left[\frac{3}{4},\ -y,\ y\right]$	[10,50]
11	$\left[y+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}-y\right]$	[11,118]
12	$\left[\frac{1}{2}-y,\ \frac{1}{4},\ \frac{1}{2}-y\right]$	[12,120]
13	$\left[\frac{1}{4},\ y+\frac{1}{2},\ y+\frac{1}{2}\right]$	[13,49]
14	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ \frac{1}{2}-y\right]$	[14,56]
15	$\left[y+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	[15,167]
16	$\left[\frac{1}{2}-y,\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	[16,165]

continued ...

Table 8

No.	position	mapping
17	$\left[y,\ \frac{1}{4},\ y\right]$	[17,103]
18	$\left[y,\ y,\ \frac{1}{4}\right]$	[18,148]
19	$\left[-y,\ y,\ \frac{3}{4}\right]$	[19,149]
20	$\left[-y,\ \frac{3}{4},\ y\right]$	[20,102]
21	$\left[-y,\ -y,\ \frac{1}{4}\right]$	[21,160]
22	$\left[y,\ \frac{3}{4},\ -y\right]$	[22,107]
23	$\left[y,\ -y,\ \frac{3}{4}\right]$	[23,159]
24	$\left[-y,\ \frac{1}{4},\ -y\right]$	[24,108]
25	$\left[\frac{3}{4},\ -y,\ -y\right]$	[25,85]
26	$\left[\frac{1}{4},\ y+\frac{1}{2},\ \frac{1}{2}-y\right]$	[26,82]
27	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ y+\frac{1}{2}\right]$	[27,81]
28	$\left[\frac{1}{2}-y,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	[28,186]
29	$\left[y+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	[29,187]
30	$\left[y+\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}-y\right]$	[30,140]
31	$\left[\frac{1}{2}-y,\ \frac{3}{4},\ \frac{1}{2}-y\right]$	[31,137]
32	$\left[\frac{3}{4},\ y,\ y\right]$	[32,86]
33	$\left[\frac{1}{4},\ -y,\ y\right]$	[33,75]
34	$\left[\frac{1}{4},\ y,\ -y\right]$	[34,74]
35	$\left[\frac{1}{2}-y,\ \frac{1}{4},\ y+\frac{1}{2}\right]$	[35,142]
36	$\left[y+\frac{1}{2},\ \frac{3}{4},\ y+\frac{1}{2}\right]$	[36,144]
37	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ \frac{1}{2}-y\right]$	[37,73]
38	$\left[\frac{3}{4},\ y+\frac{1}{2},\ y+\frac{1}{2}\right]$	[38,80]
39	$\left[\frac{1}{2}-y,\ y+\frac{1}{2},\ \frac{1}{4}\right]$	[39,191]
40	$\left[y+\frac{1}{2},\ y+\frac{1}{2},\ \frac{3}{4}\right]$	[40,189]
41	$\left[-y,\ \frac{3}{4},\ -y\right]$	[41,127]
42	$\left[-y,\ -y,\ \frac{3}{4}\right]$	[42,172]
43	$\left[y,\ -y,\ \frac{1}{4}\right]$	[43,173]
44	$\left[y,\ \frac{1}{4},\ -y\right]$	[44,126]
45	$\left[y,\ y,\ \frac{3}{4}\right]$	[45,184]
46	$\left[-y,\ \frac{1}{4},\ y\right]$	[46,131]
47	$\left[-y,\ y,\ \frac{1}{4}\right]$	[47,183]
48	$\left[y,\ \frac{3}{4},\ y\right]$	[48,132]
49	$\left[y+\frac{1}{2},\ y,\ \frac{3}{4}\right]$	[52,114]
50	$\left[\frac{1}{2}-y,\ y,\ \frac{1}{4}\right]$	[53,115]
51	$\left[\frac{1}{2}-y,\ \frac{1}{4},\ y\right]$	[54,164]
52	$\left[y+\frac{1}{2},\ \frac{3}{4},\ y\right]$	[55,161]
53	$\left[y+\frac{1}{2},\ \frac{1}{4},\ -y\right]$	[59,166]
54	$\left[\frac{1}{2}-y,\ \frac{3}{4},\ -y\right]$	[60,168]
55	$\left[y+\frac{1}{2},\ -y,\ \frac{1}{4}\right]$	[63,119]
56	$\left[\frac{1}{2}-y,\ -y,\ \frac{3}{4}\right]$	[64,117]
57	$\left[y,\ \frac{3}{4},\ y+\frac{1}{2}\right]$	[65,151]
58	$\left[y,\ y+\frac{1}{2},\ \frac{3}{4}\right]$	[66,100]
59	$\left[-y,\ y+\frac{1}{2},\ \frac{1}{4}\right]$	[67,101]
60	$\left[-y,\ \frac{1}{4},\ y+\frac{1}{2}\right]$	[68,150]
61	$\left[-y,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	[69,112]
62	$\left[y,\ \frac{1}{4},\ \frac{1}{2}-y\right]$	[70,155]
63	$\left[y,\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	[71,111]

continued ...

Table 8

No.	position	mapping
64	$\left[-y,\ \frac{3}{4},\ \frac{1}{2}-y\right]$	[72,156]
65	$\left[\frac{1}{2}-y,\ -y,\ \frac{1}{4}\right]$	[76,138]
66	$\left[y+\frac{1}{2},\ -y,\ \frac{3}{4}\right]$	[77,139]
67	$\left[y+\frac{1}{2},\ \frac{3}{4},\ -y\right]$	[78,188]
68	$\left[\frac{1}{2}-y,\ \frac{1}{4},\ -y\right]$	[79,185]
69	$\left[\frac{1}{2}-y,\ \frac{3}{4},\ y\right]$	[83,190]
70	$\left[y+\frac{1}{2},\ \frac{1}{4},\ y\right]$	[84,192]
71	$\left[\frac{1}{2}-y,\ y,\ \frac{3}{4}\right]$	[87,143]
72	$\left[y+\frac{1}{2},\ y,\ \frac{1}{4}\right]$	[88,141]
73	$\left[-y,\ \frac{1}{4},\ \frac{1}{2}-y\right]$	[89,175]
74	$\left[-y,\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	[90,124]
75	$\left[y,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	[91,125]
76	$\left[y,\ \frac{3}{4},\ \frac{1}{2}-y\right]$	[92,174]
77	$\left[y,\ y+\frac{1}{2},\ \frac{1}{4}\right]$	[93,136]
78	$\left[-y,\ \frac{3}{4},\ y+\frac{1}{2}\right]$	[94,179]
79	$\left[-y,\ y+\frac{1}{2},\ \frac{3}{4}\right]$	[95,135]
80	$\left[y,\ \frac{1}{4},\ y+\frac{1}{2}\right]$	[96,180]
81	$\left[\frac{3}{4},\ y,\ y+\frac{1}{2}\right]$	[97,157]
82	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ y\right]$	[98,154]
83	$\left[\frac{1}{4},\ y+\frac{1}{2},\ -y\right]$	[99,153]
84	$\left[\frac{3}{4},\ -y,\ \frac{1}{2}-y\right]$	[104,158]
85	$\left[\frac{1}{4},\ y,\ \frac{1}{2}-y\right]$	[105,147]
86	$\left[\frac{1}{4},\ -y,\ y+\frac{1}{2}\right]$	[106,146]
87	$\left[\frac{3}{4},\ y+\frac{1}{2},\ y\right]$	[109,145]
88	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ -y\right]$	[110,152]
89	$\left[\frac{1}{4},\ -y,\ \frac{1}{2}-y\right]$	[121,181]
90	$\left[\frac{3}{4},\ y+\frac{1}{2},\ -y\right]$	[122,178]
91	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ y\right]$	[123,177]
92	$\left[\frac{1}{4},\ y,\ y+\frac{1}{2}\right]$	[128,182]
93	$\left[\frac{3}{4},\ -y,\ y+\frac{1}{2}\right]$	[129,171]
94	$\left[\frac{3}{4},\ y,\ \frac{1}{2}-y\right]$	[130,170]
95	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ -y\right]$	[133,169]
96	$\left[\frac{1}{4},\ y+\frac{1}{2},\ y\right]$	[134,176]

Table 9: Wyckoff site: 96i, site symmetry: m..

No.	position	mapping
1	$[0,\ y,\ z]$	[1,32]
2	$\left[\frac{1}{2},\ \frac{1}{2}-z,\ y+\frac{1}{2}\right]$	[2,27]
3	$\left[\frac{1}{2},\ z+\frac{1}{2},\ \frac{1}{2}-y\right]$	[3,26]
4	$\left[z+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[4,40]
5	$\left[\frac{1}{2}-z,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[5,39]
6	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[6,35]
7	$\left[y+\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[7,36]
8	$[0,\ -y,\ -z]$	[8,25]

continued ...

Table 9

No.	position	mapping
9	$[0,\ y,\ -z]$	[9,34]
10	$[0,\ -y,\ z]$	[10,33]
11	$\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[11,30]
12	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[12,31]
13	$\left[\frac{1}{2},\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$	[13,38]
14	$\left[\frac{1}{2},\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$	[14,37]
15	$\left[z+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[15,29]
16	$\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[16,28]
17	$[z,\ 0,\ y]$	[17,48]
18	$[y,\ z,\ 0]$	[18,45]
19	$[-y,\ z,\ 0]$	[19,47]
20	$[-z,\ 0,\ y]$	[20,46]
21	$[-y,\ -z,\ 0]$	[21,42]
22	$[z,\ 0,\ -y]$	[22,44]
23	$[y,\ -z,\ 0]$	[23,43]
24	$[-z,\ 0,\ -y]$	[24,41]
25	$\left[0,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[49,80]
26	$\left[\frac{1}{2},\ -z,\ y\right]$	[50,75]
27	$\left[\frac{1}{2},\ z,\ -y\right]$	[51,74]
28	$\left[z+\frac{1}{2},\ y,\ 0\right]$	[52,88]
29	$\left[\frac{1}{2}-z,\ y,\ 0\right]$	[53,87]
30	$\left[\frac{1}{2}-y,\ 0,\ z\right]$	[54,83]
31	$\left[y+\frac{1}{2},\ 0,\ z\right]$	[55,84]
32	$\left[0,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[56,73]
33	$\left[0,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[57,82]
34	$\left[0,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[58,81]
35	$\left[y+\frac{1}{2},\ 0,\ -z\right]$	[59,78]
36	$\left[\frac{1}{2}-y,\ 0,\ -z\right]$	[60,79]
37	$\left[\frac{1}{2},\ z,\ y\right]$	[61,86]
38	$\left[\frac{1}{2},\ -z,\ -y\right]$	[62,85]
39	$\left[z+\frac{1}{2},\ -y,\ 0\right]$	[63,77]
40	$\left[\frac{1}{2}-z,\ -y,\ 0\right]$	[64,76]
41	$\left[z,\ \frac{1}{2},\ y+\frac{1}{2}\right]$	[65,96]
42	$\left[y,\ z+\frac{1}{2},\ \frac{1}{2}\right]$	[66,93]
43	$\left[-y,\ z+\frac{1}{2},\ \frac{1}{2}\right]$	[67,95]
44	$\left[-z,\ \frac{1}{2},\ y+\frac{1}{2}\right]$	[68,94]
45	$\left[-y,\ \frac{1}{2}-z,\ \frac{1}{2}\right]$	[69,90]
46	$\left[z,\ \frac{1}{2},\ \frac{1}{2}-y\right]$	[70,92]
47	$\left[y,\ \frac{1}{2}-z,\ \frac{1}{2}\right]$	[71,91]
48	$\left[-z,\ \frac{1}{2},\ \frac{1}{2}-y\right]$	[72,89]
49	$\left[\frac{1}{2},\ y,\ z+\frac{1}{2}\right]$	[97,128]
50	$\left[0,\ \frac{1}{2}-z,\ y\right]$	[98,123]
51	$\left[0,\ z+\frac{1}{2},\ -y\right]$	[99,122]
52	$\left[z,\ y+\frac{1}{2},\ 0\right]$	[100,136]
53	$\left[-z,\ y+\frac{1}{2},\ 0\right]$	[101,135]
54	$\left[-y,\ \frac{1}{2},\ z\right]$	[102,131]
55	$\left[y,\ \frac{1}{2},\ z\right]$	[103,132]

continued ...

Table 9

No.	position	mapping
56	$\left[\frac{1}{2}, -y, \frac{1}{2}-z\right]$	[104,121]
57	$\left[\frac{1}{2}, y, \frac{1}{2}-z\right]$	[105,130]
58	$\left[\frac{1}{2}, -y, z+\frac{1}{2}\right]$	[106,129]
59	$\left[y, \frac{1}{2}, -z\right]$	[107,126]
60	$\left[-y, \frac{1}{2}, -z\right]$	[108,127]
61	$\left[0, z+\frac{1}{2}, y\right]$	[109,134]
62	$\left[0, \frac{1}{2}-z, -y\right]$	[110,133]
63	$\left[z, \frac{1}{2}-y, 0\right]$	[111,125]
64	$\left[-z, \frac{1}{2}-y, 0\right]$	[112,124]
65	$\left[z+\frac{1}{2}, 0, y+\frac{1}{2}\right]$	[113,144]
66	$\left[y+\frac{1}{2}, z, \frac{1}{2}\right]$	[114,141]
67	$\left[\frac{1}{2}-y, z, \frac{1}{2}\right]$	[115,143]
68	$\left[\frac{1}{2}-z, 0, y+\frac{1}{2}\right]$	[116,142]
69	$\left[\frac{1}{2}-y, -z, \frac{1}{2}\right]$	[117,138]
70	$\left[z+\frac{1}{2}, 0, \frac{1}{2}-y\right]$	[118,140]
71	$\left[y+\frac{1}{2}, -z, \frac{1}{2}\right]$	[119,139]
72	$\left[\frac{1}{2}-z, 0, \frac{1}{2}-y\right]$	[120,137]
73	$\left[\frac{1}{2}, y+\frac{1}{2}, z\right]$	[145,176]
74	$\left[0, -z, y+\frac{1}{2}\right]$	[146,171]
75	$\left[0, z, \frac{1}{2}-y\right]$	[147,170]
76	$\left[z, y, \frac{1}{2}\right]$	[148,184]
77	$\left[-z, y, \frac{1}{2}\right]$	[149,183]
78	$\left[-y, 0, z+\frac{1}{2}\right]$	[150,179]
79	$\left[y, 0, z+\frac{1}{2}\right]$	[151,180]
80	$\left[\frac{1}{2}, \frac{1}{2}-y, -z\right]$	[152,169]
81	$\left[\frac{1}{2}, y+\frac{1}{2}, -z\right]$	[153,178]
82	$\left[\frac{1}{2}, \frac{1}{2}-y, z\right]$	[154,177]
83	$\left[y, 0, \frac{1}{2}-z\right]$	[155,174]
84	$\left[-y, 0, \frac{1}{2}-z\right]$	[156,175]
85	$\left[0, z, y+\frac{1}{2}\right]$	[157,182]
86	$\left[0, -z, \frac{1}{2}-y\right]$	[158,181]
87	$\left[z, -y, \frac{1}{2}\right]$	[159,173]
88	$\left[-z, -y, \frac{1}{2}\right]$	[160,172]
89	$\left[z+\frac{1}{2}, \frac{1}{2}, y\right]$	[161,192]
90	$\left[y+\frac{1}{2}, z+\frac{1}{2}, 0\right]$	[162,189]
91	$\left[\frac{1}{2}-y, z+\frac{1}{2}, 0\right]$	[163,191]
92	$\left[\frac{1}{2}-z, \frac{1}{2}, y\right]$	[164,190]
93	$\left[\frac{1}{2}-y, \frac{1}{2}-z, 0\right]$	[165,186]
94	$\left[z+\frac{1}{2}, \frac{1}{2}, -y\right]$	[166,188]
95	$\left[y+\frac{1}{2}, \frac{1}{2}-z, 0\right]$	[167,187]
96	$\left[\frac{1}{2}-z, \frac{1}{2}, -y\right]$	[168,185]

Table 10: Wyckoff site: `192j`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[x+\frac{1}{2},\ \frac{1}{2}-z,\ y+\frac{1}{2}\right]$	`[2]`
3	$\left[x+\frac{1}{2},\ z+\frac{1}{2},\ \frac{1}{2}-y\right]$	`[3]`
4	$\left[z+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[4]`
5	$\left[\frac{1}{2}-z,\ y+\frac{1}{2},\ x+\frac{1}{2}\right]$	`[5]`
6	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[6]`
7	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[7]`
8	$[x,\ -y,\ -z]$	`[8]`
9	$[-x,\ y,\ -z]$	`[9]`
10	$[-x,\ -y,\ z]$	`[10]`
11	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[11]`
12	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	`[12]`
13	$\left[\frac{1}{2}-x,\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$	`[13]`
14	$\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$	`[14]`
15	$\left[z+\frac{1}{2},\ \frac{1}{2}-y,\ x+\frac{1}{2}\right]$	`[15]`
16	$\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ \frac{1}{2}-x\right]$	`[16]`
17	$[z,\ x,\ y]$	`[17]`
18	$[y,\ z,\ x]$	`[18]`
19	$[-y,\ z,\ -x]$	`[19]`
20	$[-z,\ -x,\ y]$	`[20]`
21	$[-y,\ -z,\ x]$	`[21]`
22	$[z,\ -x,\ -y]$	`[22]`
23	$[y,\ -z,\ -x]$	`[23]`
24	$[-z,\ x,\ -y]$	`[24]`
25	$[-x,\ -y,\ -z]$	`[25]`
26	$\left[\frac{1}{2}-x,\ z+\frac{1}{2},\ \frac{1}{2}-y\right]$	`[26]`
27	$\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ y+\frac{1}{2}\right]$	`[27]`
28	$\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ x+\frac{1}{2}\right]$	`[28]`
29	$\left[z+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-x\right]$	`[29]`
30	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	`[30]`
31	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[31]`
32	$[-x,\ y,\ z]$	`[32]`
33	$[x,\ -y,\ z]$	`[33]`
34	$[x,\ y,\ -z]$	`[34]`
35	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[35]`
36	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[36]`
37	$\left[x+\frac{1}{2},\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$	`[37]`
38	$\left[x+\frac{1}{2},\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$	`[38]`
39	$\left[\frac{1}{2}-z,\ y+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[39]`
40	$\left[z+\frac{1}{2},\ y+\frac{1}{2},\ x+\frac{1}{2}\right]$	`[40]`
41	$[-z,\ -x,\ -y]$	`[41]`
42	$[-y,\ -z,\ -x]$	`[42]`
43	$[y,\ -z,\ x]$	`[43]`
44	$[z,\ x,\ -y]$	`[44]`
45	$[y,\ z,\ -x]$	`[45]`
46	$[-z,\ x,\ y]$	`[46]`

continued ...

Table 10

No.	position	mapping
47	$[-y,\, z,\, x]$	[47]
48	$[z,\, -x,\, y]$	[48]
49	$\left[x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[49]
50	$\left[x+\frac{1}{2},\, -z,\, y\right]$	[50]
51	$\left[x+\frac{1}{2},\, z,\, -y\right]$	[51]
52	$\left[z+\frac{1}{2},\, y,\, -x\right]$	[52]
53	$\left[\frac{1}{2}-z,\, y,\, x\right]$	[53]
54	$\left[\frac{1}{2}-y,\, x,\, z\right]$	[54]
55	$\left[y+\frac{1}{2},\, -x,\, z\right]$	[55]
56	$\left[x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[56]
57	$\left[-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[57]
58	$\left[-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[58]
59	$\left[y+\frac{1}{2},\, x,\, -z\right]$	[59]
60	$\left[\frac{1}{2}-y,\, -x,\, -z\right]$	[60]
61	$\left[\frac{1}{2}-x,\, z,\, y\right]$	[61]
62	$\left[\frac{1}{2}-x,\, -z,\, -y\right]$	[62]
63	$\left[z+\frac{1}{2},\, -y,\, x\right]$	[63]
64	$\left[\frac{1}{2}-z,\, -y,\, -x\right]$	[64]
65	$\left[z,\, x+\frac{1}{2},\, y+\frac{1}{2}\right]$	[65]
66	$\left[y,\, z+\frac{1}{2},\, x+\frac{1}{2}\right]$	[66]
67	$\left[-y,\, z+\frac{1}{2},\, \frac{1}{2}-x\right]$	[67]
68	$\left[-z,\, \frac{1}{2}-x,\, y+\frac{1}{2}\right]$	[68]
69	$\left[-y,\, \frac{1}{2}-z,\, x+\frac{1}{2}\right]$	[69]
70	$\left[z,\, \frac{1}{2}-x,\, \frac{1}{2}-y\right]$	[70]
71	$\left[y,\, \frac{1}{2}-z,\, \frac{1}{2}-x\right]$	[71]
72	$\left[-z,\, x+\frac{1}{2},\, \frac{1}{2}-y\right]$	[72]
73	$\left[-x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[73]
74	$\left[\frac{1}{2}-x,\, z,\, -y\right]$	[74]
75	$\left[\frac{1}{2}-x,\, -z,\, y\right]$	[75]
76	$\left[\frac{1}{2}-z,\, -y,\, x\right]$	[76]
77	$\left[z+\frac{1}{2},\, -y,\, -x\right]$	[77]
78	$\left[y+\frac{1}{2},\, -x,\, -z\right]$	[78]
79	$\left[\frac{1}{2}-y,\, x,\, -z\right]$	[79]
80	$\left[-x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[80]
81	$\left[x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[81]
82	$\left[x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[82]
83	$\left[\frac{1}{2}-y,\, -x,\, z\right]$	[83]
84	$\left[y+\frac{1}{2},\, x,\, z\right]$	[84]
85	$\left[x+\frac{1}{2},\, -z,\, -y\right]$	[85]
86	$\left[x+\frac{1}{2},\, z,\, y\right]$	[86]
87	$\left[\frac{1}{2}-z,\, y,\, -x\right]$	[87]
88	$\left[z+\frac{1}{2},\, y,\, x\right]$	[88]
89	$\left[-z,\, \frac{1}{2}-x,\, \frac{1}{2}-y\right]$	[89]
90	$\left[-y,\, \frac{1}{2}-z,\, \frac{1}{2}-x\right]$	[90]
91	$\left[y,\, \frac{1}{2}-z,\, x+\frac{1}{2}\right]$	[91]
92	$\left[z,\, x+\frac{1}{2},\, \frac{1}{2}-y\right]$	[92]
93	$\left[y,\, z+\frac{1}{2},\, \frac{1}{2}-x\right]$	[93]

continued ...

Table 10

No.	position	mapping
94	$\left[-z,\, x+\frac{1}{2},\, y+\frac{1}{2}\right]$	[94]
95	$\left[-y,\, z+\frac{1}{2},\, x+\frac{1}{2}\right]$	[95]
96	$\left[z,\, \frac{1}{2}-x,\, y+\frac{1}{2}\right]$	[96]
97	$\left[x+\frac{1}{2},\, y,\, z+\frac{1}{2}\right]$	[97]
98	$\left[x,\, \frac{1}{2}-z,\, y\right]$	[98]
99	$\left[x,\, z+\frac{1}{2},\, -y\right]$	[99]
100	$\left[z,\, y+\frac{1}{2},\, -x\right]$	[100]
101	$\left[-z,\, y+\frac{1}{2},\, x\right]$	[101]
102	$\left[-y,\, x+\frac{1}{2},\, z\right]$	[102]
103	$\left[y,\, \frac{1}{2}-x,\, z\right]$	[103]
104	$\left[x+\frac{1}{2},\, -y,\, \frac{1}{2}-z\right]$	[104]
105	$\left[\frac{1}{2}-x,\, y,\, \frac{1}{2}-z\right]$	[105]
106	$\left[\frac{1}{2}-x,\, -y,\, z+\frac{1}{2}\right]$	[106]
107	$\left[y,\, x+\frac{1}{2},\, -z\right]$	[107]
108	$\left[-y,\, \frac{1}{2}-x,\, -z\right]$	[108]
109	$\left[-x,\, z+\frac{1}{2},\, y\right]$	[109]
110	$\left[-x,\, \frac{1}{2}-z,\, -y\right]$	[110]
111	$\left[z,\, \frac{1}{2}-y,\, x\right]$	[111]
112	$\left[-z,\, \frac{1}{2}-y,\, -x\right]$	[112]
113	$\left[z+\frac{1}{2},\, x,\, y+\frac{1}{2}\right]$	[113]
114	$\left[y+\frac{1}{2},\, z,\, x+\frac{1}{2}\right]$	[114]
115	$\left[\frac{1}{2}-y,\, z,\, \frac{1}{2}-x\right]$	[115]
116	$\left[\frac{1}{2}-z,\, -x,\, y+\frac{1}{2}\right]$	[116]
117	$\left[\frac{1}{2}-y,\, -z,\, x+\frac{1}{2}\right]$	[117]
118	$\left[z+\frac{1}{2},\, -x,\, \frac{1}{2}-y\right]$	[118]
119	$\left[y+\frac{1}{2},\, -z,\, \frac{1}{2}-x\right]$	[119]
120	$\left[\frac{1}{2}-z,\, x,\, \frac{1}{2}-y\right]$	[120]
121	$\left[\frac{1}{2}-x,\, -y,\, \frac{1}{2}-z\right]$	[121]
122	$\left[-x,\, z+\frac{1}{2},\, -y\right]$	[122]
123	$\left[-x,\, \frac{1}{2}-z,\, y\right]$	[123]
124	$\left[-z,\, \frac{1}{2}-y,\, x\right]$	[124]
125	$\left[z,\, \frac{1}{2}-y,\, -x\right]$	[125]
126	$\left[y,\, \frac{1}{2}-x,\, -z\right]$	[126]
127	$\left[-y,\, x+\frac{1}{2},\, -z\right]$	[127]
128	$\left[\frac{1}{2}-x,\, y,\, z+\frac{1}{2}\right]$	[128]
129	$\left[x+\frac{1}{2},\, -y,\, z+\frac{1}{2}\right]$	[129]
130	$\left[x+\frac{1}{2},\, y,\, \frac{1}{2}-z\right]$	[130]
131	$\left[-y,\, \frac{1}{2}-x,\, z\right]$	[131]
132	$\left[y,\, x+\frac{1}{2},\, z\right]$	[132]
133	$\left[x,\, \frac{1}{2}-z,\, -y\right]$	[133]
134	$\left[x,\, z+\frac{1}{2},\, y\right]$	[134]
135	$\left[-z,\, y+\frac{1}{2},\, -x\right]$	[135]
136	$\left[z,\, y+\frac{1}{2},\, x\right]$	[136]
137	$\left[\frac{1}{2}-z,\, -x,\, \frac{1}{2}-y\right]$	[137]
138	$\left[\frac{1}{2}-y,\, -z,\, \frac{1}{2}-x\right]$	[138]
139	$\left[y+\frac{1}{2},\, -z,\, x+\frac{1}{2}\right]$	[139]
140	$\left[z+\frac{1}{2},\, x,\, \frac{1}{2}-y\right]$	[140]

continued ...

Table 10

No.	position	mapping
141	$\left[y+\frac{1}{2},\, z,\, \frac{1}{2}-x\right]$	[141]
142	$\left[\frac{1}{2}-z,\, x,\, y+\frac{1}{2}\right]$	[142]
143	$\left[\frac{1}{2}-y,\, z,\, x+\frac{1}{2}\right]$	[143]
144	$\left[z+\frac{1}{2},\, -x,\, y+\frac{1}{2}\right]$	[144]
145	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	[145]
146	$\left[x,\, -z,\, y+\frac{1}{2}\right]$	[146]
147	$\left[x,\, z,\, \frac{1}{2}-y\right]$	[147]
148	$\left[z,\, y,\, \frac{1}{2}-x\right]$	[148]
149	$\left[-z,\, y,\, x+\frac{1}{2}\right]$	[149]
150	$\left[-y,\, x,\, z+\frac{1}{2}\right]$	[150]
151	$\left[y,\, -x,\, z+\frac{1}{2}\right]$	[151]
152	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	[152]
153	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, -z\right]$	[153]
154	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	[154]
155	$\left[y,\, x,\, \frac{1}{2}-z\right]$	[155]
156	$\left[-y,\, -x,\, \frac{1}{2}-z\right]$	[156]
157	$\left[-x,\, z,\, y+\frac{1}{2}\right]$	[157]
158	$\left[-x,\, -z,\, \frac{1}{2}-y\right]$	[158]
159	$\left[z,\, -y,\, x+\frac{1}{2}\right]$	[159]
160	$\left[-z,\, -y,\, \frac{1}{2}-x\right]$	[160]
161	$\left[z+\frac{1}{2},\, x+\frac{1}{2},\, y\right]$	[161]
162	$\left[y+\frac{1}{2},\, z+\frac{1}{2},\, x\right]$	[162]
163	$\left[\frac{1}{2}-y,\, z+\frac{1}{2},\, -x\right]$	[163]
164	$\left[\frac{1}{2}-z,\, \frac{1}{2}-x,\, y\right]$	[164]
165	$\left[\frac{1}{2}-y,\, \frac{1}{2}-z,\, x\right]$	[165]
166	$\left[z+\frac{1}{2},\, \frac{1}{2}-x,\, -y\right]$	[166]
167	$\left[y+\frac{1}{2},\, \frac{1}{2}-z,\, -x\right]$	[167]
168	$\left[\frac{1}{2}-z,\, x+\frac{1}{2},\, -y\right]$	[168]
169	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, -z\right]$	[169]
170	$\left[-x,\, z,\, \frac{1}{2}-y\right]$	[170]
171	$\left[-x,\, -z,\, y+\frac{1}{2}\right]$	[171]
172	$\left[-z,\, -y,\, x+\frac{1}{2}\right]$	[172]
173	$\left[z,\, -y,\, \frac{1}{2}-x\right]$	[173]
174	$\left[y,\, -x,\, \frac{1}{2}-z\right]$	[174]
175	$\left[-y,\, x,\, \frac{1}{2}-z\right]$	[175]
176	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	[176]
177	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z\right]$	[177]
178	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$	[178]
179	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	[179]
180	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[180]
181	$\left[x,\, -z,\, \frac{1}{2}-y\right]$	[181]
182	$\left[x,\, z,\, y+\frac{1}{2}\right]$	[182]
183	$\left[-z,\, y,\, \frac{1}{2}-x\right]$	[183]
184	$\left[z,\, y,\, x+\frac{1}{2}\right]$	[184]
185	$\left[\frac{1}{2}-z,\, \frac{1}{2}-x,\, -y\right]$	[185]
186	$\left[\frac{1}{2}-y,\, \frac{1}{2}-z,\, -x\right]$	[186]
187	$\left[y+\frac{1}{2},\, \frac{1}{2}-z,\, x\right]$	[187]

continued ...

Table 10

No.	position	mapping
188	$\left[z+\frac{1}{2},\, x+\frac{1}{2},\, -y\right]$	[188]
189	$\left[y+\frac{1}{2},\, z+\frac{1}{2},\, -x\right]$	[189]
190	$\left[\frac{1}{2}-z,\, x+\frac{1}{2},\, y\right]$	[190]
191	$\left[\frac{1}{2}-y,\, z+\frac{1}{2},\, x\right]$	[191]
192	$\left[z+\frac{1}{2},\, \frac{1}{2}-x,\, y\right]$	[192]